HEALTHCARE OPERATIONAL ANALYTICS · CASE STUDY

# Hospital readmission penalty risk

A Power BI tool that shows hospitals where fixing readmissions saves the most money

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DOMAIN
Healthcare operations / CMS HRRP

TOOLS
Power BI, DAX, data modeling

SCALE
2,833 hospitals

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> Medicare's readmission program can cut a hospital's payments by up to 3% if too many patients come back within 30 days. I pulled together readmission, quality, and ownership data for 2,833 hospitals and built a Power BI tool that shows each hospital how much money is at stake. The point is simple: instead of just telling hospitals their readmissions are high, it shows them where fixing the problem pays off the most. It turned out that 83% of hospitals are already in penalty range, with the biggest problems in Massachusetts and in hip and knee surgeries.

## 01 The problem

Medicare runs a program called the Hospital Readmissions Reduction Program, or HRRP. If a hospital sends too many patients home and they end up back in the hospital within 30 days, Medicare cuts that hospital's payments. This applies to common conditions like heart failure, pneumonia, COPD, and hip and knee replacements. The penalty can go as high as 3% of everything Medicare pays that hospital, which is a lot of money for any hospital to lose.

Here is the gap I wanted to close. Hospitals can already see which of their readmission rates are high. What they usually cannot see is **where spending money to improve would save them the most in penalties**. Most readmission dashboards just point at the problem and stop. I wanted this one to connect the medical numbers to the money decision behind them.

## 02 The data and how I built it

I combined three data sources into one hospital-level view: Medicare's readmission ratios, patient experience and star rating scores, and hospital ownership data from Hospital Compare. Together that covered 2,833 hospitals across the country.

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|---|---|---|---|
| **2,833** HOSPITALS | **2,358** AT-RISK (83%) | **MA** WORST STATE | **1.31** HIGHEST RATIO |

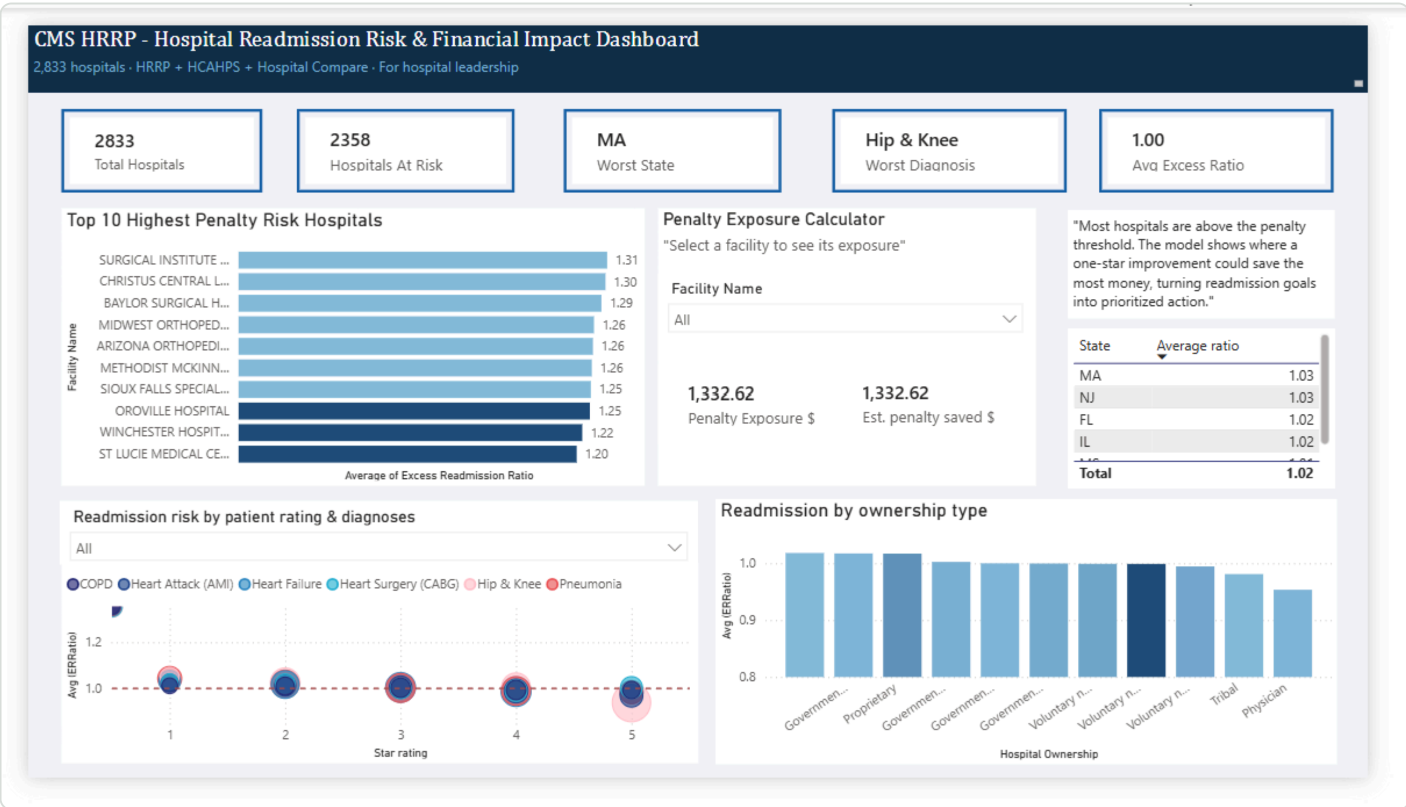
The heart of the tool is a calculator I built in DAX. It estimates how much money a hospital could lose in penalties, and how much it could save by improving its star rating by one step:

```
// How much a hospital risks losing. The more its ratio is above 1.0, the bigger the penalty.
Penalty Risk = (Excess Ratio − 1.0) × Medicare Revenue × 3%

// How much it saves from a one-star improvement. Capped so it can't over-count.
Penalty Saved = MIN(Excess Ratio − 1.0, 0.020) × Medicare Revenue × 3%
```

I used the `MIN` function for a reason. My analysis showed that improving by one star lowers the readmission ratio by about 0.020. But if a hospital is only just above the penalty line, one star of improvement would wipe out its whole penalty, and no more. The cap makes sure the tool does not give a hospital credit for saving more than it actually owes.

## 03 The dashboard




*The final dashboard: summary numbers at the top, a ranking of the riskiest hospitals, a calculator for any single hospital, and breakdowns by state, condition, and ownership type.*

I laid it out so it reads from the top down, like a decision. The summary numbers show how big the problem is. The ranking and the calculator answer who is at risk and by how much. The charts at the bottom explain why, by looking at patient ratings, medical conditions, and who owns the hospital. There is also a dropdown so a user can pick any single hospital and see its own numbers.

## 04 What I found

01 **83% of hospitals are at risk.** Out of 2,833 hospitals, 2,358 were already above the line where penalties kick in. This is a problem across the whole system, not just a few bad hospitals.

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02 **Massachusetts is the worst state.** When I ranked states by how bad their readmissions are on average, Massachusetts and New Jersey came out on top. This is different from just counting hospitals, which only tells you which states are biggest.

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03 **Hip and knee surgeries are the biggest driver.** Of all the conditions, joint replacements had the worst readmission numbers across the board.

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04 **Ownership type matters.** Physician-owned and tribal hospitals did better than government-run and for-profit hospitals.

## 05 A decision I had to think through

JUDGMENT CALL

My first version ranked states by how many at-risk hospitals they had. That put California, Texas, and Florida at the top. But when I looked closer, all that really showed was which states are big. Bigger states have more hospitals, so of course they have more at-risk hospitals. It did not tell me where readmissions were actually worst.

So I changed it to rank states by their **average readmission score** instead, and added a rule to skip states with too few hospitals so one bad hospital could not throw off the numbers. That measured how bad the problem really is, not just how big the state is, and it correctly pointed to Massachusetts. It also lined up with the "worst state" number at the top of the dashboard, so the whole page told one consistent story.

## 06 What this tool does not do

**These are estimates, not Medicare's exact numbers.** I built the penalty math on how Medicare's program actually works, but I made some simplifications on purpose:

- I used the same Medicare revenue figure for every hospital, so the differences you see come from readmission scores, not hospital size.
- I compared every hospital to one simple cutoff, while Medicare uses more detailed benchmarks for each condition and adjusts for patient mix.
- My state ranking treats every hospital equally. A fuller version would give bigger hospitals more weight.

A real production version would pull each hospital's actual Medicare revenue, use Medicare's full benchmark method, and weight hospitals by size. That would turn this from a tool that describes the problem into one that makes recommendations for each hospital.

## 07 Why it matters

The real value here is helping hospitals decide where to spend first. Instead of treating "reduce readmissions" as one big goal, this tool shows a hospital leader **exactly where improving by one star turns into real money saved**. That lets them put a limited budget where it pays off most, and compare the risk across different hospitals, conditions, and ownership types using the same dollar figure.